The Future Belongs to These Open Source DevOps Tools

Organisations are adopting DevOps today for their own good. This article briefly discusses a few open source DevOps tools that will be used a lot in the near future.

DevOps is a culture, and implementing its practices helps manage the application life cycle in an effective manner using automation. It has three pillars — people, processes, and tools. Tools play a critical role in DevOps or culture transformation initiatives. People and processes help in maintaining uniformity and sustainability across business units or projects in an organisation that aims for faster time to market with high quality for end users.

Category-wise list of open source DevOps tools

Category	Tools
Source code management	GitLab, Git, GitHub, Atlassian BitBucket,
Code analysis tools	SonarQube
CI/CD	Jenkins, Microsoft Azure DevOps, TeamCity, GitLab, GitHub, Atlassian Bamboo
Test automation	Selenium, Appium, JMeter
Configuration management	Ansible, Puppet, Chef
Containerisation and orchestration	Docker and Kubernetes
Cloud service providers	Amazon Web Services, Microsoft Azure, Google Cloud Platform
Monitoring	Prometheus, Grafana
Logging	Elastic Search, FluentD, FluentBit, Kibana
Communication and collaboration	Microsoft Teams, Slack, Skype and Zoom

I have used a few open source tools and I believe that these are very important for implementing DevOps practices in 2022 and beyond. Let's take a brief look at them.

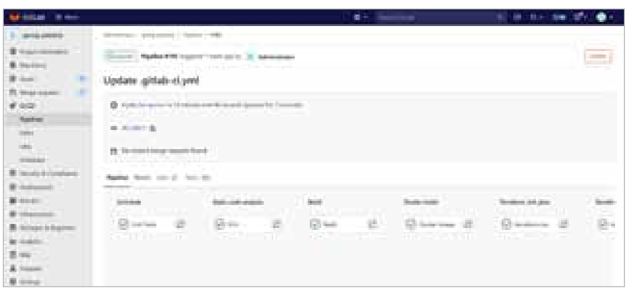

Figure 1: GitLab CI/CD pipeline

GitLab

This DevOps platform provides the ability to plan, develop, secure, and operate software in a single application. Its essential features include cache and artifact management.

In my experience, GitLab has one of the most futuristic features for implementing DevOps practices.

Initial release	2014
Stable release	14.6.0
Written in	Ruby, Go and JavaScript
Licence	Community Edition - MIT License
Website	*about.gitlab.com*
GitHub repository	*https://gitlab.com/gitlab-org/gitlab-foss/*
Features	• Highly scalable • Supports collaboration • Supports on-premise and cloud installation
How is it useful in implementing DevOps practices?	DevOps platform • Version control • CI/CD • Package management • Resource management • Security • Distributed architecture • Cloud support
Can we integrate it with Pipeline as Code?	Yes. The command is: `Gitlab-ci.yaml`
Is a commercial flavour available?	Yes
Pricing	*https://about.gitlab.com/pricing/Free, Premium, Ultimate*
Community	2500+ code contributors
Use cases	• Value stream management • Agile development • Source code management • Continuous integration (CI/CD) • Out-of-the-box pipelines • Security (DevSecOps) • GitOps • The DevOps platform